EX6 - Desenhando Emojis com Dados

Nota final

Formativa | Última tentativa com nota

Não avaliado

ENVIADO: 23/04/26 23:55

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Não avaliado

Instruções   Envio

Desafio 1: O Criador de Emojis (Pixels RGB)

```
emoji_data = {
 "nome": "Smile",
 "grade": [
  [(255,255,0), (255,255,0), (255,255,0), (255,255,0), (255,255,0)], # Linha 0
  [(255,255,0), (0,0,0),  (255,255,0), (0,0,0),  (255,255,0)], # Linha 1 (Olhos)
  [(255,255,0), (255,255,0), (255,255,0), (255,255,0), (255,255,0)], # Linha 2
  [(255,255,0), (0,0,0),  (0,0,0),  (0,0,0),  (255,255,0)], # Linha 3 (Boca)
  [(255,255,0), (255,255,0), (255,255,0), (255,255,0), (255,255,0)] # Linha 4
 ]
}
nova_grade = []
grade_original = emoji_data["grade"]
for linha in grade_original:
 nova_linha = []
 for pixel in linha:
  if pixel == (255, 255, 0):
   pixel_sombreado = (pixel[0] // 2, pixel[1] // 2, pixel[2] // 2)
   nova_linha.append(pixel_sombreado)
  else:
   nova_linha.append(pixel)
 nova_grade.append(nova_linha)
emoji_data["grade"] = nova_grade
for linha in emoji_data["grade"]:
 print(linha)
plt.imshow(emoji_data["grade"])
plt.show()
```

Desafio 2: Matrizes Musicais (Transposição)

```
biblioteca_musical = {
 "Toques": [
  {"Alegre": ([440, 480], [520, 560])},
  {"Triste": ([200, 150], [100, 50])}
 ]
}
for item in biblioteca_musical["Toques"]:
 for nome_musica, matriz in list(item.items()):
```

```
    nova_matriz = (
      [matriz[0][0], matriz[1][0]],
      [matriz[0][1], matriz[1][1]]
    )
    item.update({nome_musica: nova_matriz})
print("Biblioteca:")
for toque in biblioteca_musical["Toques"]:
  print(toque)
```

Desafio 3: O Integrador

```
playlist_imagens = {
  "vibe": [
    [(255, 0, 0), (0, 255, 0)],
    [(0, 0, 255), (255, 255, 0)],
    [(255, 255, 255), (0, 0, 0)]
  ]
}
print(f"Playlist: {list(playlist_imagens.keys())[0]}")
for nome_video, frames in playlist_imagens.items():
  frame_removido = frames.pop()
  print(f"{nome_video}: {frame_removido}")

for i in range(len(frames)):
   linha_atual = frames[i]
   nova_linha = []
   for pixel in linha_atual:
      if pixel.count(255) >= 1:
        pixel_processado = (pixel[0] // 2, pixel[1] // 2, pixel[2] // 2)
        nova_linha.append(pixel_processado)
      else:
        nova_linha.append(pixel)
   frames.pop(i)
   frames.insert(i, nova_linha)

for frame in playlist_imagens["vibe"]:
  print(frame)
plt.imshow(playlist_imagens["vibe"])
plt.show()
```